

Oracle HR Schema Subquery

Analytics & Business Reporting

1

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME, SALARY  
FROM EMPLOYEES WHERE SALARY > (SELECT AVG(SALARY)  
FROM EMPLOYEES);
```

- Logic: compares employee salary with company average.

-Type: scalar subquery.



The screenshot shows a 'Query Result' window with a toolbar containing icons for a pin, print, refresh, and close, along with an 'SQL' label. Below the toolbar, it states 'Fetched 50 rows in 0.133 seconds'. The table below displays the results of the query, with columns for EMPLOYEE_ID, FULL_NAME, and SALARY. The data is sorted by salary in descending order.

	EMPLOYEE_ID	FULL_NAME	SALARY
1	100	Steven King	24000
2	101	Neena Kochhar	17000
3	102	Lex De Haan	17000
4	103	Alexander Hunold	9000
5	108	Nancy Greenberg	12008
6	109	Daniel Faviyet	9000
7	110	John Chen	8200
8	111	Ismael Sciarra	7700
9	112	Jose Manuel Urman	7800
10	113	Luis Popp	6900
11	114	Den Raphaely	11000
12	120	Matthew Weiss	8000
13	121	Adam Fripp	8200
14	122	Payam Kaufling	7900
15	123	Shanta Vollman	6500

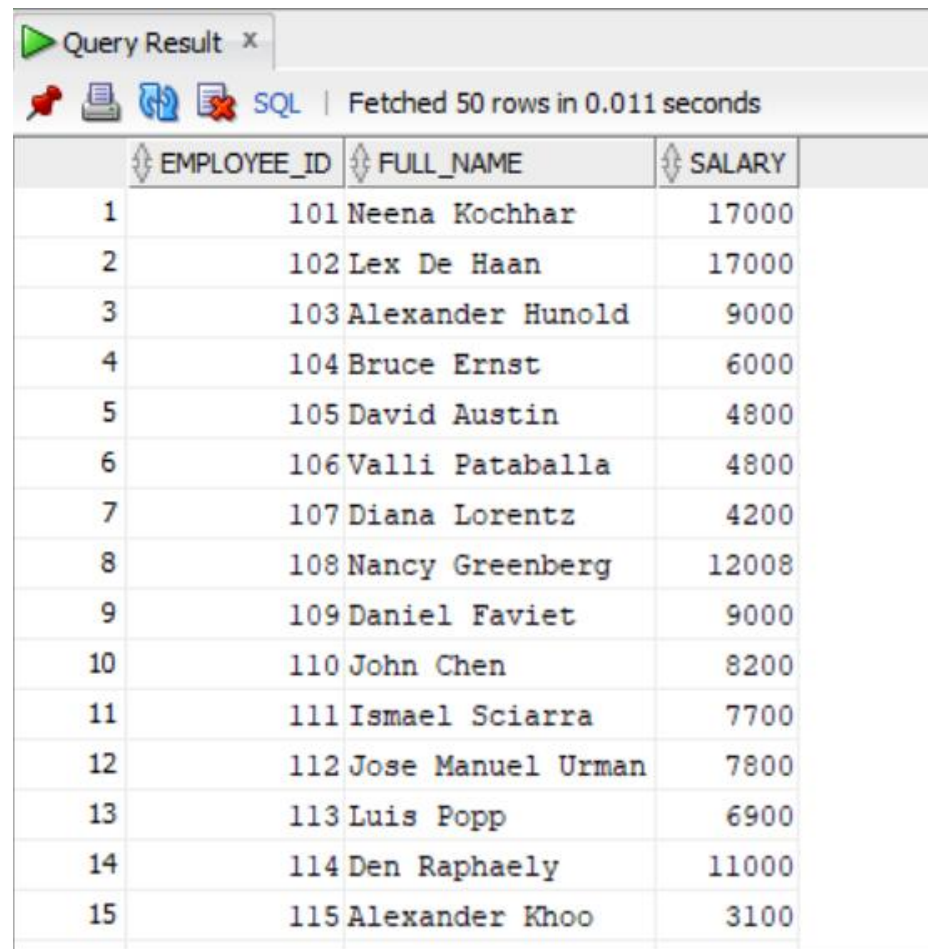
2

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME, SALARY  
FROM EMPLOYEES
```

```
WHERE SALARY < (SELECT MAX(SALARY) FROM EMPLOYEES);
```

- Logic: shows employees earning less than the highest salary.

- Type: scalar subquery.



Query Result x

SQL | Fetched 50 rows in 0.011 seconds

	EMPLOYEE_ID	FULL_NAME	SALARY
1	101	Neena Kochhar	17000
2	102	Lex De Haan	17000
3	103	Alexander Hunold	9000
4	104	Bruce Ernst	6000
5	105	David Austin	4800
6	106	Valli Pataballa	4800
7	107	Diana Lorentz	4200
8	108	Nancy Greenberg	12008
9	109	Daniel Faviet	9000
10	110	John Chen	8200
11	111	Ismael Sciarra	7700
12	112	Jose Manuel Urman	7800
13	113	Luis Popp	6900
14	114	Den Raphaely	11000
15	115	Alexander Khoo	3100

3

```
SELECT DEPARTMENT_ID, AVG(SALARY) AS AVG_SALARY
FROM EMPLOYEES
GROUP BY DEPARTMENT_ID
HAVING AVG(SALARY) > (SELECT AVG(SALARY) FROM EMPLOYEES);
```

- **Logic:** compares department average salary with company average.

-Type: scalar subquery with GROUP BY.

[illegible]

4

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME,  
DEPARTMENT_ID, SALARY  
  
FROM EMPLOYEES E WHERE SALARY = (SELECT MAX(SALARY) FROM  
EMPLOYEES WHERE DEPARTMENT_ID = E.DEPARTMENT_ID);
```

-Logic: finds highest paid employee in each department.

-Type: correlated subquery.

EMPLOYEE_ID	FULL_NAME	DEPARTMENT_ID	SALARY
1	100 Steven King	90	24000
2	103 Alexander Hunold	60	9000
3	108 Nancy Greenberg	100	12008
4	114 Den Raphaely	30	11000
5	121 Adam Fripp	50	8200
6	145 John Russell	80	14000
7	200 Jennifer Whalen	10	4400
8	201 Michael Hartstein	20	13000
9	203 Susan Mavris	40	6500
10	204 Hermann Baer	70	10000
11	205 Shelley Higgins	110	12008

5

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME,  
DEPARTMENT_ID FROM EMPLOYEES WHERE DEPARTMENT_ID IN (  
SELECT DEPARTMENT_ID FROM DEPARTMENTS WHERE LOCATION_ID IN (  
SELECT LOCATION_ID FROM LOCATIONS WHERE COUNTRY_ID = 'US'));
```

-Logic: finds employees in departments located in United States.

-Type: nested multi-row subquery.

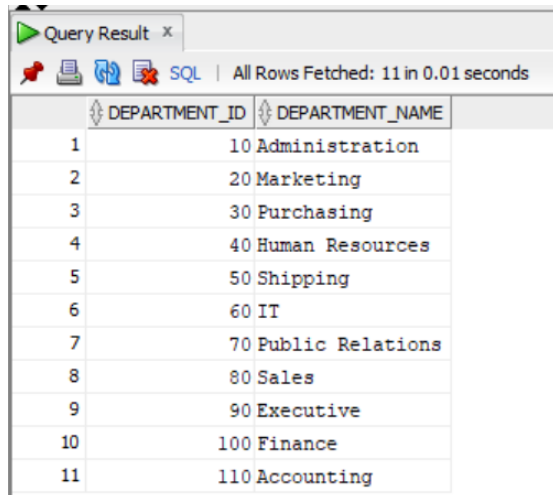
Query Result x			
SQL Fetched 50 rows in 0.044 seconds			
	EMPLOYEE_ID	FULL_NAME	DEPARTMENT_ID
1	103	Alexander Hunold	60
2	104	Bruce Ernst	60
3	105	David Austin	60
4	106	Valli Pataballa	60
5	107	Diana Lorentz	60
6	120	Matthew Weiss	50
7	121	Adam Fripp	50
8	122	Payam Kaufling	50
9	123	Shanta Vollman	50
10	124	Kevin Mourgous	50
11	125	Julia Nayer	50
12	126	Irene Mikkilineni	50
13	127	James Landry	50
14	128	Steven Markle	50
15	129	Laura Bissot	50
16	130	Mozhe Atkinson	50
17	131	James Marlow	50
18	132	TJ Olson	50

6

```
SELECT DEPARTMENT_ID, DEPARTMENT_NAME FROM DEPARTMENTS  
WHERE DEPARTMENT_ID IN ( SELECT DISTINCT DEPARTMENT_ID FROM  
EMPLOYEES WHERE DEPARTMENT_ID IS NOT NULL);
```

-Logic: shows departments that have employees.

-Type: multi-row subquery.



Query Result x

SQL | All Rows Fetched: 11 in 0.01 seconds

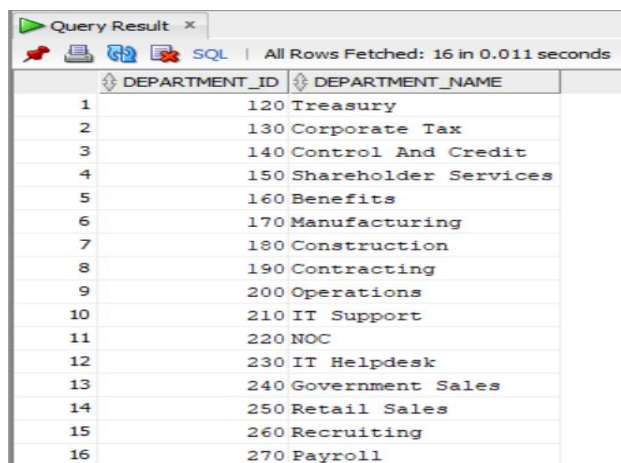
	DEPARTMENT_ID	DEPARTMENT_NAME
1	10	Administration
2	20	Marketing
3	30	Purchasing
4	40	Human Resources
5	50	Shipping
6	60	IT
7	70	Public Relations
8	80	Sales
9	90	Executive
10	100	Finance
11	110	Accounting

7

```
SELECT DEPARTMENT_ID, DEPARTMENT_NAME FROM DEPARTMENTS WHERE  
DEPARTMENT_ID NOT IN ( SELECT DISTINCT DEPARTMENT_ID FROM  
EMPLOYEES WHERE DEPARTMENT_ID IS NOT NULL);
```

-Logic: shows departments with no employees.

-Type: multi-row subquery.



Query Result x

SQL | All Rows Fetched: 16 in 0.011 seconds

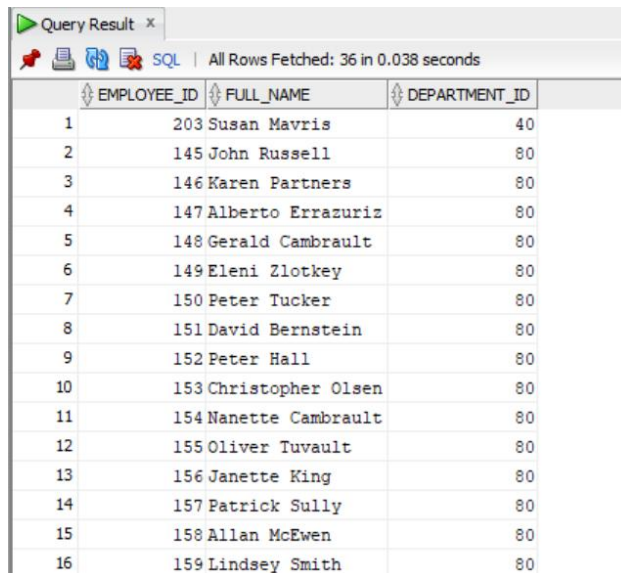
	DEPARTMENT_ID	DEPARTMENT_NAME
1	120	Treasury
2	130	Corporate Tax
3	140	Control And Credit
4	150	Shareholder Services
5	160	Benefits
6	170	Manufacturing
7	180	Construction
8	190	Contracting
9	200	Operations
10	210	IT Support
11	220	NOC
12	230	IT Helpdesk
13	240	Government Sales
14	250	Retail Sales
15	260	Recruiting
16	270	Payroll

8

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME,  
DEPARTMENT_ID FROM EMPLOYEES WHERE DEPARTMENT_ID IN (  
  
SELECT DEPARTMENT_ID FROM DEPARTMENTS WHERE LOCATION_ID IN (  
  
SELECT LOCATION_ID FROM LOCATIONS WHERE COUNTRY_ID IN (  
  
SELECT COUNTRY_ID FROM COUNTRIES WHERE REGION_ID = (  
  
SELECT REGION_ID FROM REGIONS WHERE REGION_NAME = 'Europe'))));
```

-Logic: finds employees working in departments located in Europe.

-Type: nested subquery.



Query Result x

SQL | All Rows Fetched: 36 in 0.038 seconds

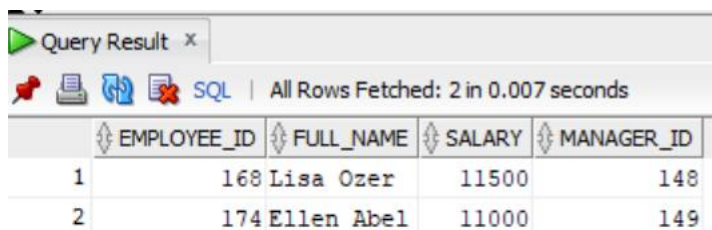
	EMPLOYEE_ID	FULL_NAME	DEPARTMENT_ID
1	203	Susan Mavris	40
2	145	John Russell	80
3	146	Karen Partners	80
4	147	Alberto Errazuriz	80
5	148	Gerald Cambrault	80
6	149	Eleni Zlotkey	80
7	150	Peter Tucker	80
8	151	David Bernstein	80
9	152	Peter Hall	80
10	153	Christopher Olsen	80
11	154	Nanette Cambrault	80
12	155	Oliver Tuvault	80
13	156	Janette King	80
14	157	Patrick Sully	80
15	158	Allan McEwen	80
16	159	Lindsey Smith	80

9

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME, SALARY,  
MANAGER_ID FROM EMPLOYEES E WHERE SALARY > (SELECT SALARY FROM  
EMPLOYEES M WHERE M.EMPLOYEE_ID = E.MANAGER_ID);
```

- Logic: compares employee salary with manager salary.

-Type: correlated subquery.



Query Result x

SQL | All Rows Fetched: 2 in 0.007 seconds

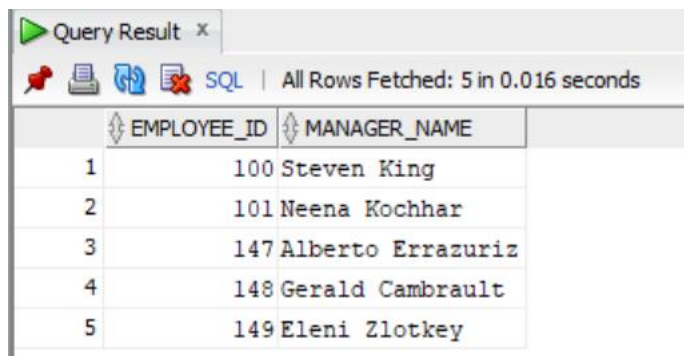
	EMPLOYEE_ID	FULL_NAME	SALARY	MANAGER_ID
1	168	Lisa Ozer	11500	148
2	174	Ellen Abel	11000	149

10

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS MANAGER_NAME  
FROM EMPLOYEES WHERE EMPLOYEE_ID IN ( SELECT MANAGER_ID FROM  
EMPLOYEES WHERE SALARY > 10000);
```

-Logic: shows managers who supervise employees earning more than 10000.

-Type: multi-row subquery.



Query Result x

SQL | All Rows Fetched: 5 in 0.016 seconds

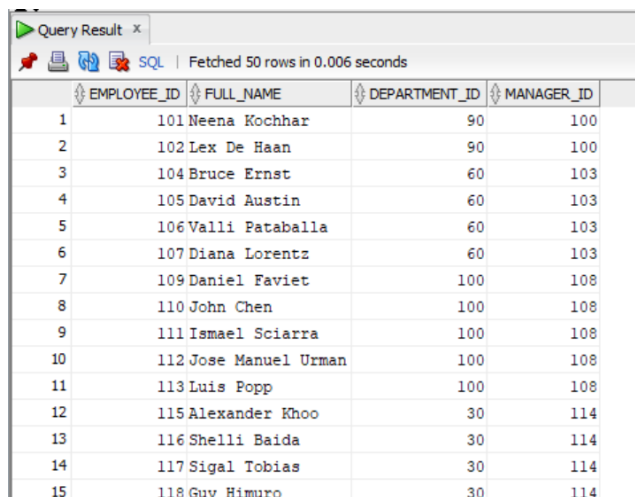
EMPLOYEE_ID	MANAGER_NAME
1	100 Steven King
2	101 Neena Kochhar
3	147 Alberto Errazuriz
4	148 Gerald Cambrault
5	149 Eleni Zlotkey

11

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME,  
DEPARTMENT_ID, MANAGER_ID FROM EMPLOYEES E  
WHERE DEPARTMENT_ID = ( SELECT DEPARTMENT_ID FROM EMPLOYEES M  
WHERE M.EMPLOYEE_ID = E.MANAGER_ID);
```

- Logic: shows employees whose manager works in same department.

- Type: correlated subquery.



Query Result x

SQL | Fetched 50 rows in 0.006 seconds

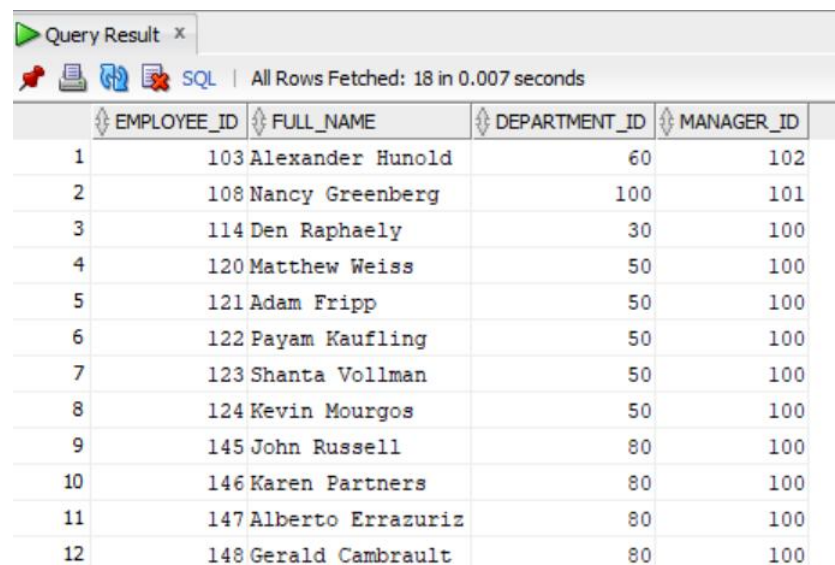
EMPLOYEE_ID	FULL_NAME	DEPARTMENT_ID	MANAGER_ID
1	101 Neena Kochhar	90	100
2	102 Lex De Haan	90	100
3	104 Bruce Ernst	60	103
4	105 David Austin	60	103
5	106 Valli Pataballa	60	103
6	107 Diana Lorentz	60	103
7	109 Daniel Faviet	100	108
8	110 John Chen	100	108
9	111 Ismael Sciarra	100	108
10	112 Jose Manuel Urman	100	108
11	113 Luis Popp	100	108
12	115 Alexander Khoo	30	114
13	116 Shelli Baida	30	114
14	117 Sigal Tobias	30	114
15	118 Guy Himuro	30	114

12

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME,  
DEPARTMENT_ID, MANAGER_ID FROM EMPLOYEES E  
  
WHERE DEPARTMENT_ID <> ( SELECT DEPARTMENT_ID FROM EMPLOYEES M  
WHERE M.EMPLOYEE_ID = E.MANAGER_ID);
```

- **Logic: shows employees whose manager works in different department.**

- **Type: correlated subquery.**



Query Result x

All Rows Fetched: 18 in 0.007 seconds

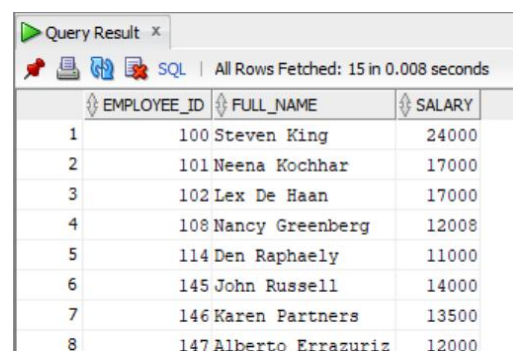
	EMPLOYEE_ID	FULL_NAME	DEPARTMENT_ID	MANAGER_ID
1	103	Alexander Hunold	60	102
2	108	Nancy Greenberg	100	101
3	114	Den Raphaely	30	100
4	120	Matthew Weiss	50	100
5	121	Adam Fripp	50	100
6	122	Payam Kaufling	50	100
7	123	Shanta Vollman	50	100
8	124	Kevin Mourgous	50	100
9	145	John Russell	80	100
10	146	Karen Partners	80	100
11	147	Alberto Errazuriz	80	100
12	148	Gerald Cambrault	80	100

13

```
SELECT * FROM (SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS  
FULL_NAME, SALARY FROM EMPLOYEES) WHERE SALARY > 10000;
```

- **Logic: inline view filters employees earning more than 10000.**

- **Type: inline view.**



Query Result x

All Rows Fetched: 15 in 0.008 seconds

	EMPLOYEE_ID	FULL_NAME	SALARY
1	100	Steven King	24000
2	101	Neena Kochhar	17000
3	102	Lex De Haan	17000
4	108	Nancy Greenberg	12008
5	114	Den Raphaely	11000
6	145	John Russell	14000
7	146	Karen Partners	13500
8	147	Alberto Errazuriz	12000

14

```
SELECT * FROM (SELECT DEPARTMENT_ID, AVG(SALARY) AS AVG_SALARY FROM
EMPLOYEES GROUP BY DEPARTMENT_ID) WHERE AVG_SALARY > 9000;
```

- Logic: inline view filters departments with average salary greater than 9000.

- Type: inline view.

[illegible]

15

```
SELECT * FROM (SELECT DEPARTMENT_ID, COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEES GROUP BY DEPARTMENT_ID) WHERE TOTAL_EMPLOYEES >
5;
```

- Logic: inline view filters departments with more than five employees.

- Type: inline view.

Query Result x

SQL | All Rows Fetched: 4 in 0.005 seconds

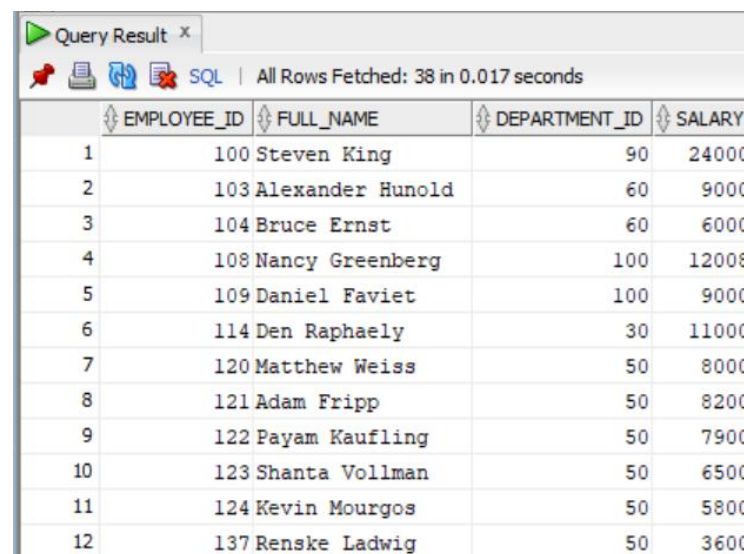
	DEPARTMENT_ID	TOTAL_EMPLOYEES
1	50	45
2	30	6
3	100	6
4	80	34

16

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME,
DEPARTMENT_ID, SALARY FROM EMPLOYEES E WHERE SALARY > ( SELECT
AVG(SALARY) FROM EMPLOYEES WHERE DEPARTMENT_ID =
E.DEPARTMENT_ID);
```

- Logic: compares employee salary with department average salary.

- Type: correlated subquery.



Query Result x

SQL | All Rows Fetched: 38 in 0.017 seconds

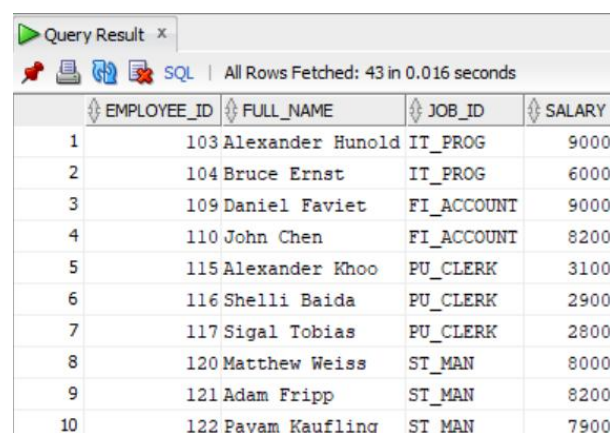
	EMPLOYEE_ID	FULL_NAME	DEPARTMENT_ID	SALARY
1	100	Steven King	90	24000
2	103	Alexander Hunold	60	9000
3	104	Bruce Ernst	60	6000
4	108	Nancy Greenberg	100	12008
5	109	Daniel Faviet	100	9000
6	114	Den Raphaely	30	11000
7	120	Matthew Weiss	50	8000
8	121	Adam Fripp	50	8200
9	122	Payam Kaufling	50	7900
10	123	Shanta Vollman	50	6500
11	124	Kevin Mourgous	50	5800
12	137	Renske Ladwig	50	3600

17

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME, JOB_ID,
SALARY FROM EMPLOYEES E WHERE SALARY > ( SELECT AVG(SALARY) FROM
EMPLOYEES WHERE JOB_ID = E.JOB_ID);
```

- Logic: compares employee salary with average salary of same job role.

- Type: correlated subquery.



Query Result x

SQL | All Rows Fetched: 43 in 0.016 seconds

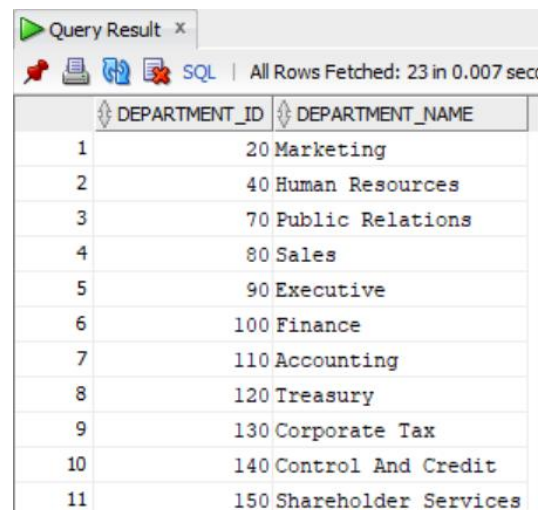
	EMPLOYEE_ID	FULL_NAME	JOB_ID	SALARY
1	103	Alexander Hunold	IT_PROG	9000
2	104	Bruce Ernst	IT_PROG	6000
3	109	Daniel Faviet	FI_ACCOUNT	9000
4	110	John Chen	FI_ACCOUNT	8200
5	115	Alexander Khoo	PU_CLERK	3100
6	116	Shelli Baida	PU_CLERK	2900
7	117	Sigal Tobias	PU_CLERK	2800
8	120	Matthew Weiss	ST_MAN	8000
9	121	Adam Fripp	ST_MAN	8200
10	122	Payam Kaufling	ST_MAN	7900

18

```
SELECT DEPARTMENT_ID, DEPARTMENT_NAME FROM DEPARTMENTS D  
WHERE NOT EXISTS ( SELECT 1 FROM EMPLOYEES E WHERE E.DEPARTMENT_ID  
= D.DEPARTMENT_ID AND E.SALARY <= 5000);
```

- Logic: shows departments where no employee earns 5000 or less.

- Type: correlated subquery with NOT EXISTS.



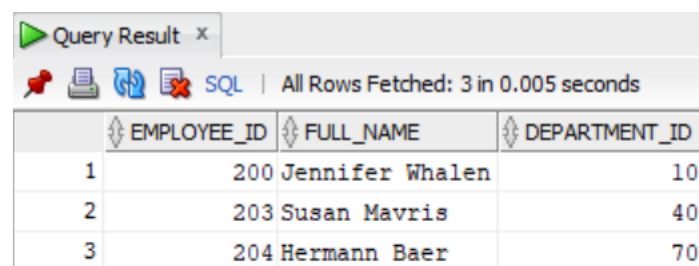
	DEPARTMENT_ID	DEPARTMENT_NAME
1	20	Marketing
2	40	Human Resources
3	70	Public Relations
4	80	Sales
5	90	Executive
6	100	Finance
7	110	Accounting
8	120	Treasury
9	130	Corporate Tax
10	140	Control And Credit
11	150	Shareholder Services

19

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME,  
DEPARTMENT_ID FROM EMPLOYEES E WHERE 1 = ( SELECT COUNT(*) FROM  
EMPLOYEES WHERE DEPARTMENT_ID = E.DEPARTMENT_ID);
```

- Logic: shows employees who are the only employee in their department.

- Type: correlated subquery.



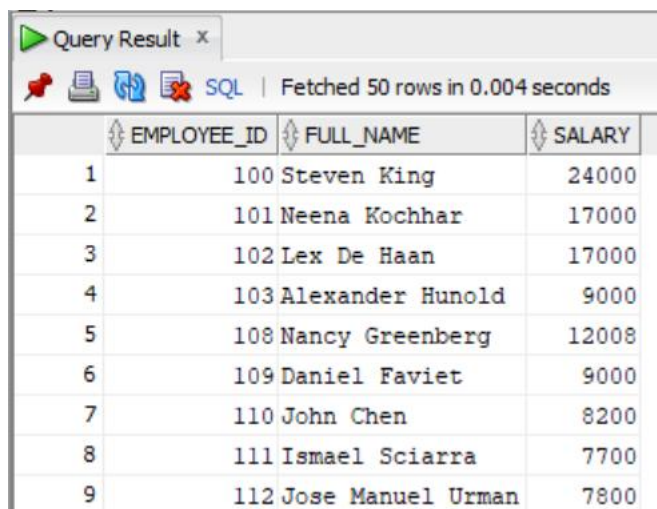
	EMPLOYEE_ID	FULL_NAME	DEPARTMENT_ID
1	200	Jennifer Whalen	10
2	203	Susan Mavris	40
3	204	Hermann Baer	70

20

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME, SALARY  
FROM EMPLOYEES  
WHERE SALARY > (SELECT AVG(SALARY) FROM EMPLOYEES);
```

-Logic: shows employees earning above company average salary.

-Type: scalar subquery.



Query Result x

SQL | Fetched 50 rows in 0.004 seconds

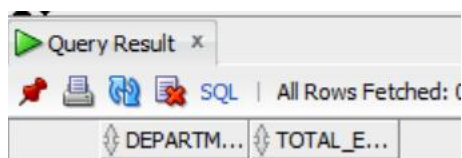
	EMPLOYEE_ID	FULL_NAME	SALARY
1	100	Steven King	24000
2	101	Neena Kochhar	17000
3	102	Lex De Haan	17000
4	103	Alexander Hunold	9000
5	108	Nancy Greenberg	12008
6	109	Daniel Faviet	9000
7	110	John Chen	8200
8	111	Ismael Sciarra	7700
9	112	Jose Manuel Urman	7800

21

```
SELECT DEPARTMENT_ID, COUNT(*) AS TOTAL_EMPLOYEES  
FROM EMPLOYEES GROUP BY DEPARTMENT_ID HAVING COUNT(*) > (  
SELECT COUNT(*) FROM EMPLOYEES WHERE DEPARTMENT_ID = 50);
```

- Logic: compares each department employee count with Department 50.

- Type: scalar subquery with HAVING.



Query Result x

SQL | All Rows Fetched: 0

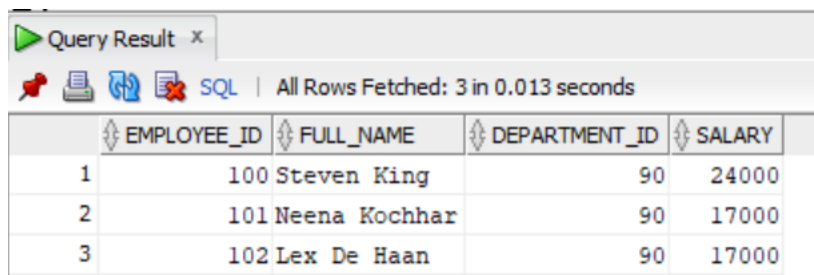
DEPARTM...	TOTAL_E...
------------	------------

22

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME,  
DEPARTMENT_ID, SALARY FROM EMPLOYEES WHERE DEPARTMENT_ID = (  
  
SELECT DEPARTMENT_ID FROM EMPLOYEES GROUP BY DEPARTMENT_ID  
  
HAVING AVG(SALARY) = ( SELECT MAX(AVG(SALARY)) FROM EMPLOYEES  
GROUP BY DEPARTMENT_ID));
```

- Logic: finds employees in the department with highest average salary.

- Type: nested subquery.



Query Result x

All Rows Fetched: 3 in 0.013 seconds

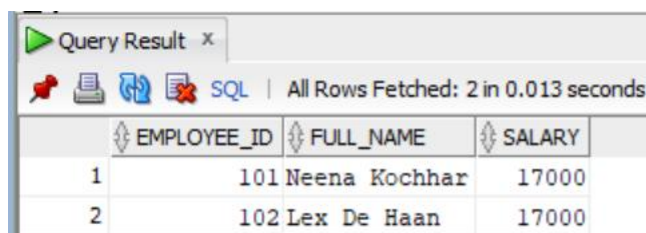
	EMPLOYEE_ID	FULL_NAME	DEPARTMENT_ID	SALARY
1	100	Steven King	90	24000
2	101	Neena Kochhar	90	17000
3	102	Lex De Haan	90	17000

23

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS FULL_NAME, SALARY  
FROM EMPLOYEES WHERE SALARY = ( SELECT MAX(SALARY) FROM EMPLOYEES  
WHERE SALARY < (SELECT MAX(SALARY) FROM EMPLOYEES));
```

-Logic: finds employees earning the second-highest salary.

-Type: nested scalar subquery.



Query Result x

All Rows Fetched: 2 in 0.013 seconds

	EMPLOYEE_ID	FULL_NAME	SALARY
1	101	Neena Kochhar	17000
2	102	Lex De Haan	17000

24

```
SELECT EMPLOYEE_ID, FIRST_NAME || ' ' || LAST_NAME AS MANAGER_NAME  
FROM EMPLOYEES WHERE EMPLOYEE_ID IN ( SELECT MANAGER_ID FROM  
EMPLOYEES GROUP BY MANAGER_ID HAVING AVG(SALARY) > 10000);
```

- **Logic: shows managers whose employees average salary exceeds 10000.**

- **Type: multi-row subquery with GROUP BY.**



The screenshot shows a 'Query Result' window with a toolbar containing icons for a pin, print, refresh, and error, along with a 'SQL' label and the text 'All Rows Fetched: 1 in 0.013 seconds'. Below the toolbar is a table with two columns: 'EMPLOYEE_ID' and 'MANAGER_NAME'. The table contains one row with the values '1' and '100 Steven King'.

EMPLOYEE_ID	MANAGER_NAME
1	100 Steven King